NBA Defensive Analysis & Player Recommendation System

Complete SQL Syntax / Queries

This document contains the readable SQL queries and view definitions used for the NBA defensive analysis project.

1. View Available Tables

```
USE nba_defense;
SHOW TABLES;
```

2. View Team Defensive Data

```
SELECT *
FROM team_defense;
```

3. Compare Selected Teams

```
SELECT
    team_name,
    dreb,
    stl,
    blk,
    def_rtg,
    dreb_pct,
    opp_pts_off_tov,
    opp_pts_2nd_chance,
    opp_pts_fb,
    opp_pts_paint
FROM team_defense
WHERE team_name IN ('LAL', 'BKN', 'GSW', 'DEN');
```

4. Check a Specific Team

```
SELECT *
FROM team_defense
WHERE team_name = 'BKN';
```

5. View Player Defensive Statistics

```
SELECT *
FROM defensive_stats
LIMIT 5;
```

6. Compare Selected Players

```
SELECT
    player_name,
    team_name,
    dreb_per_game,
    stl_per_game,
```

```
    blk_per_game
FROM defensive_stats
WHERE player_name IN (
    'Nikola Jokić',
    'Walker Kessler',
    'Anthony Davis',
    'Scottie Barnes'
);
```

7. Player Defensive Strength

```
SELECT
    player_name,
    team_name,
    dreb_per_game,
    stl_per_game,
    blk_per_game,
    rebounding_strength,
    perimeter_strength,
    rim_strength
FROM player_defensive_strength
ORDER BY player_name;
```

8. Identify Team Defensive Needs

```
SELECT *
FROM team_defensive_needs;
```

9. View Defensive Recommendations

```
SELECT *
FROM defensive_recommendations
ORDER BY target_team, match_score DESC;
```

10. Final Player Recommendations

```
SELECT *
FROM final_defensive_recommendations
ORDER BY target_team, match_score DESC;
```

11. Show Recommendations With Salary Under $25 Million

```
SELECT
    target_team,
    player_name,
    current_team,
    salary_millions,
    match_score
FROM defensive_recommendations
WHERE salary_millions <= 25
  AND match_score > 0
ORDER BY target_team, match_score DESC;
```

12. Player Defensive Strength View

```
CREATE OR REPLACE VIEW player_defensive_strength AS
SELECT
    defensive_stats.player_name AS player_name,
    defensive_stats.team_name AS team_name,
    defensive_stats.dreb_per_game AS dreb_per_game,
    defensive_stats.stl_per_game AS stl_per_game,
    defensive_stats.blk_per_game AS blk_per_game,

    ROUND(PERCENT_RANK() OVER (
        ORDER BY defensive_stats.dreb_per_game
    ) * 100, 2) AS rebounding_strength,

    ROUND(PERCENT_RANK() OVER (
        ORDER BY defensive_stats.stl_per_game
    ) * 100, 2) AS perimeter_strength,

    ROUND(PERCENT_RANK() OVER (
        ORDER BY defensive_stats.blk_per_game
    ) * 100, 2) AS rim_strength

FROM defensive_stats
WHERE defensive_stats.dreb_per_game IS NOT NULL
  AND defensive_stats.stl_per_game IS NOT NULL
  AND defensive_stats.blk_per_game IS NOT NULL;
```

13. Team Defensive Needs View

```
CREATE OR REPLACE VIEW team_defensive_needs AS
SELECT
    team_name,

    CASE
        WHEN dreb <= (SELECT AVG(dreb) FROM team_defense)
        THEN 'Weak Rebounding'
        ELSE 'Good Rebounding'
    END AS rebounding_status,

    CASE
        WHEN stl <= (SELECT AVG(stl) FROM team_defense)
        THEN 'Weak Perimeter Defense'
        ELSE 'Good Perimeter Defense'
    END AS perimeter_status,

    CASE
        WHEN blk <= (SELECT AVG(blk) FROM team_defense)
```

```
        THEN 'Weak Rim Protection'
        ELSE 'Good Rim Protection'
    END AS rim_protection_status,

    CASE
        WHEN opp_pts_2nd_chance >=
             (SELECT AVG(opp_pts_2nd_chance) FROM team_defense)
        THEN 'Allows Too Many 2nd Chance Points'
        ELSE 'Good 2nd Chance Defense'
    END AS second_chance_status,

    CASE
        WHEN opp_pts_paint >=
             (SELECT AVG(opp_pts_paint) FROM team_defense)
        THEN 'Allows Too Many Paint Points'
        ELSE 'Good Paint Defense'
    END AS paint_status

FROM team_defense;
```

14. Defensive Recommendation View

```
CREATE OR REPLACE VIEW defensive_recommendations AS

WITH team_needs AS (
    SELECT
        team_name,
        dreb,
        stl,
        blk,
        def_rtg,
        dreb_pct,
        opp_pts_off_tov,
        opp_pts_2nd_chance,
        opp_pts_fb,
        opp_pts_paint,

        NTILE(4) OVER (ORDER BY dreb) AS dreb_weakness,
        NTILE(4) OVER (ORDER BY stl) AS stl_weakness,
        NTILE(4) OVER (ORDER BY blk) AS blk_weakness,
        NTILE(4) OVER (ORDER BY def_rtg DESC) AS def_rtg_weakness,
        NTILE(4) OVER (ORDER BY dreb_pct) AS dreb_pct_weakness,
        NTILE(4) OVER (ORDER BY opp_pts_off_tov DESC) AS turnover_weakness,
        NTILE(4) OVER (ORDER BY opp_pts_2nd_chance DESC) AS second_chance_weakness,
        NTILE(4) OVER (ORDER BY opp_pts_fb DESC) AS fast_break_weakness,
        NTILE(4) OVER (ORDER BY opp_pts_paint DESC) AS paint_weakness

    FROM team_defense
),

player_profile AS (
    SELECT
        defensive_stats.player_name,
        defensive_stats.team_name AS current_team,

        (1 - PERCENT_RANK() OVER (
            ORDER BY defensive_stats.def_rtg
        )) * 100 AS def_rtg_score,

        PERCENT_RANK() OVER (
            ORDER BY defensive_stats.dreb_per_game
        ) * 100 AS dreb_score,

        PERCENT_RANK() OVER (
            ORDER BY defensive_stats.dreb_pct
        ) * 100 AS dreb_pct_score,

        PERCENT_RANK() OVER (
            ORDER BY defensive_stats.stl_per_game
        ) * 100 AS stl_score,

        PERCENT_RANK() OVER (
            ORDER BY defensive_stats.stl_pct
        ) * 100 AS stl_pct_score,

        PERCENT_RANK() OVER (
            ORDER BY defensive_stats.blk_per_game
        ) * 100 AS blk_score,

        PERCENT_RANK() OVER (
            ORDER BY defensive_stats.blk_pct
```

```
        ) * 100 AS blk_pct_score

    FROM defensive_stats

    WHERE defensive_stats.def_rtg IS NOT NULL
      AND defensive_stats.dreb_per_game IS NOT NULL
      AND defensive_stats.stl_per_game IS NOT NULL
      AND defensive_stats.blk_per_game IS NOT NULL
      AND defensive_stats.dreb_pct IS NOT NULL
      AND defensive_stats.stl_pct IS NOT NULL
      AND defensive_stats.blk_pct IS NOT NULL
)

SELECT
    t.team_name AS target_team,
    p.player_name,
    p.current_team,
    s.salary AS salary_millions,

    ROUND(
        (
            CASE WHEN t.dreb_weakness = 1
                 THEN p.dreb_score * 0.20 ELSE 0 END

            + CASE WHEN t.dreb_pct_weakness = 1
                   THEN p.dreb_pct_score * 0.15 ELSE 0 END

            + CASE WHEN t.stl_weakness = 1
                   THEN (p.stl_score * 0.10) +
                        (p.stl_pct_score * 0.15) ELSE 0 END

            + CASE WHEN t.blk_weakness = 1
                   THEN (p.blk_score * 0.10) +
                        (p.blk_pct_score * 0.15) ELSE 0 END

            + CASE WHEN t.turnover_weakness = 1
                   THEN p.stl_pct_score * 0.05 ELSE 0 END

            + CASE WHEN t.second_chance_weakness = 1
                   THEN p.dreb_pct_score * 0.05 ELSE 0 END

            + CASE WHEN t.fast_break_weakness = 1
                   THEN p.stl_pct_score * 0.05 ELSE 0 END

            + CASE WHEN t.paint_weakness = 1
                   THEN p.blk_pct_score * 0.05 ELSE 0 END
        ), 2
    ) AS match_score

FROM team_needs t
JOIN player_profile p
    ON p.current_team <> t.team_name
LEFT JOIN salaries s
    ON p.player_name = s.player_name
WHERE s.salary IS NOT NULL
  AND s.salary > 0;
```

15. Final Defensive Recommendations View

```
CREATE OR REPLACE VIEW final_defensive_recommendations AS
```

```sql
WITH ranked_players AS (
    SELECT
        target_team,
        player_name,
        current_team,
        salary_millions,
        match_score,

        ROW_NUMBER() OVER (
            PARTITION BY target_team
            ORDER BY match_score DESC
        ) AS rank_no

    FROM defensive_recommendations

    WHERE salary_millions <= 25
      AND match_score > 0
)

SELECT
    target_team,
    player_name,
    current_team,
    salary_millions,
    match_score

FROM ranked_players
WHERE rank_no = 1;
```

16. Final Output Query

```sql
SELECT
    target_team,
    player_name,
    current_team,
    salary_millions,
    match_score
FROM final_defensive_recommendations
ORDER BY target_team, match_score DESC;
```